

Circuit Simulation Lab:30

Design of SIPO

The purpose of this experiment is to design a 4 bit SIPO. A serial-in, parallel-out shift register is similar to the serial-in, serial-out shift register in that it shifts data into internal storage elements and shifts data out at the serial-out, data-out, pin.

CODE:

```
module sipo_4bit(
input clk,
input rst,
input serial_in,
output [3:0] q
);

reg [3:0] shift_reg;

always @(posedge clk)
begin

if(rst)
    shift_reg <= 4'b0000;

else
    shift_reg <= {shift_reg[2:0],serial_in};

end

assign q = shift_reg;

Endmodule

TESTBENCH
module tb_sipo_4bit;

reg clk;
reg rst;
reg serial_in;

wire [3:0] q;
```

```

sipo_4bit dut(
.clk(clk),
.rst(rst),
.serial_in(serial_in),
.q(q)
);

always #10 clk = ~clk;

initial begin

$dumpfile("sipo.vcd");
$dumpvars(0,tb_sipo_4bit);

clk = 0;
rst = 1;
serial_in = 0;

#20;
rst = 0;

// Shift in 1011

serial_in = 1; #20;
serial_in = 0; #20;
serial_in = 1; #20;
serial_in = 1; #20;

// Extra clocks

serial_in = 0; #40;

$finish;

end

initial begin

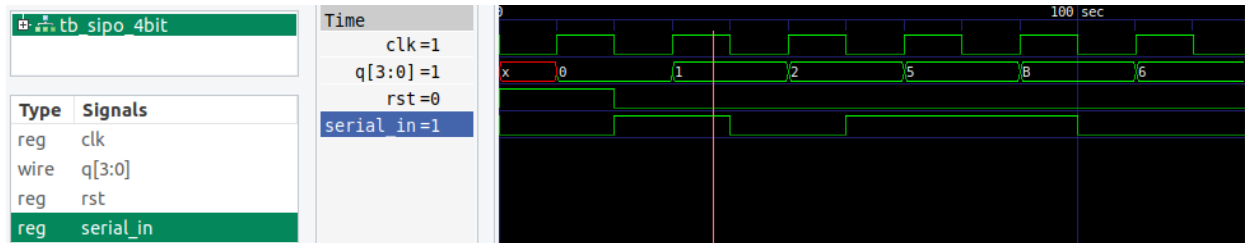
$monitor(
"time=%0t serial_in=%b q=%b",
$time,
serial_in,
q

```

);

end

Endmodule



```
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_30$ iverilog -o out
sipo_4bit.v tb_sipo_4bit.v
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_30$ vvp out
VCD info: dumpfile sipo.vcd opened for output.
time=0 serial_in=0 q=xxxx
time=10 serial_in=0 q=0000
time=20 serial_in=1 q=0000
time=30 serial_in=1 q=0001
time=40 serial_in=0 q=0001
time=50 serial_in=0 q=0010
time=60 serial_in=1 q=0010
time=70 serial_in=1 q=0101
time=90 serial_in=1 q=1011
time=100 serial_in=0 q=1011
time=110 serial_in=0 q=0110
time=130 serial_in=0 q=1100
```